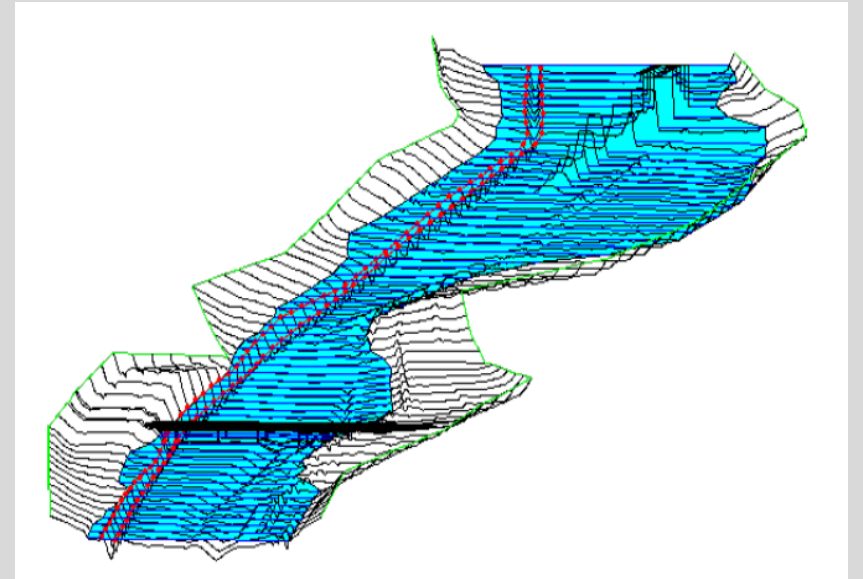


ENCN442 Surface Water and Groundwater Engineering

Surface Water Modelling

Hydraulic Modelling
- Steady -



Learning Objectives

After completion of this module the learner will be able to

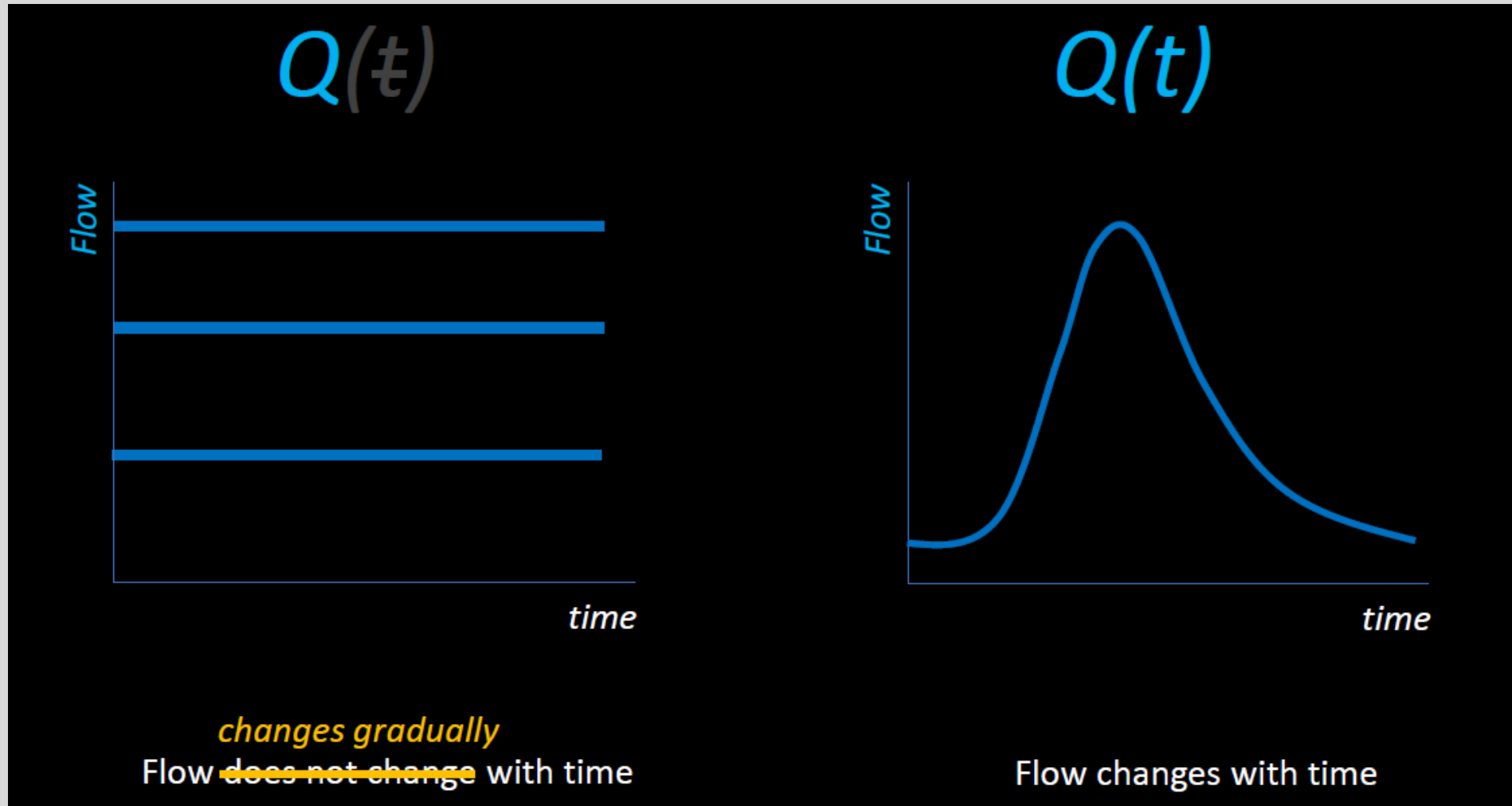
1. explain the principles and objectives of hydraulic modelling (Level 2: Understand)
2. explain the structure and variables of a hydraulic model (Level 2: Understand)
3. explain the main types of hydraulic models (Level 2: Understand)
4. explain the steps to develop a hydraulic model (Level 2: Understand) (CL)
5. develop a simple hydraulic model (Level 6: Create) (CL)
6. select model parameters (Level 5: Evaluate) (CL)
7. develop a hydraulic model (Level 6: Create) to compare and contrast the effect of implementation of various infrastructure measures on flow properties (Level 3: Apply and Level 4: Analyse)

Engineering Applications of Hydraulic Modelling (surface water)

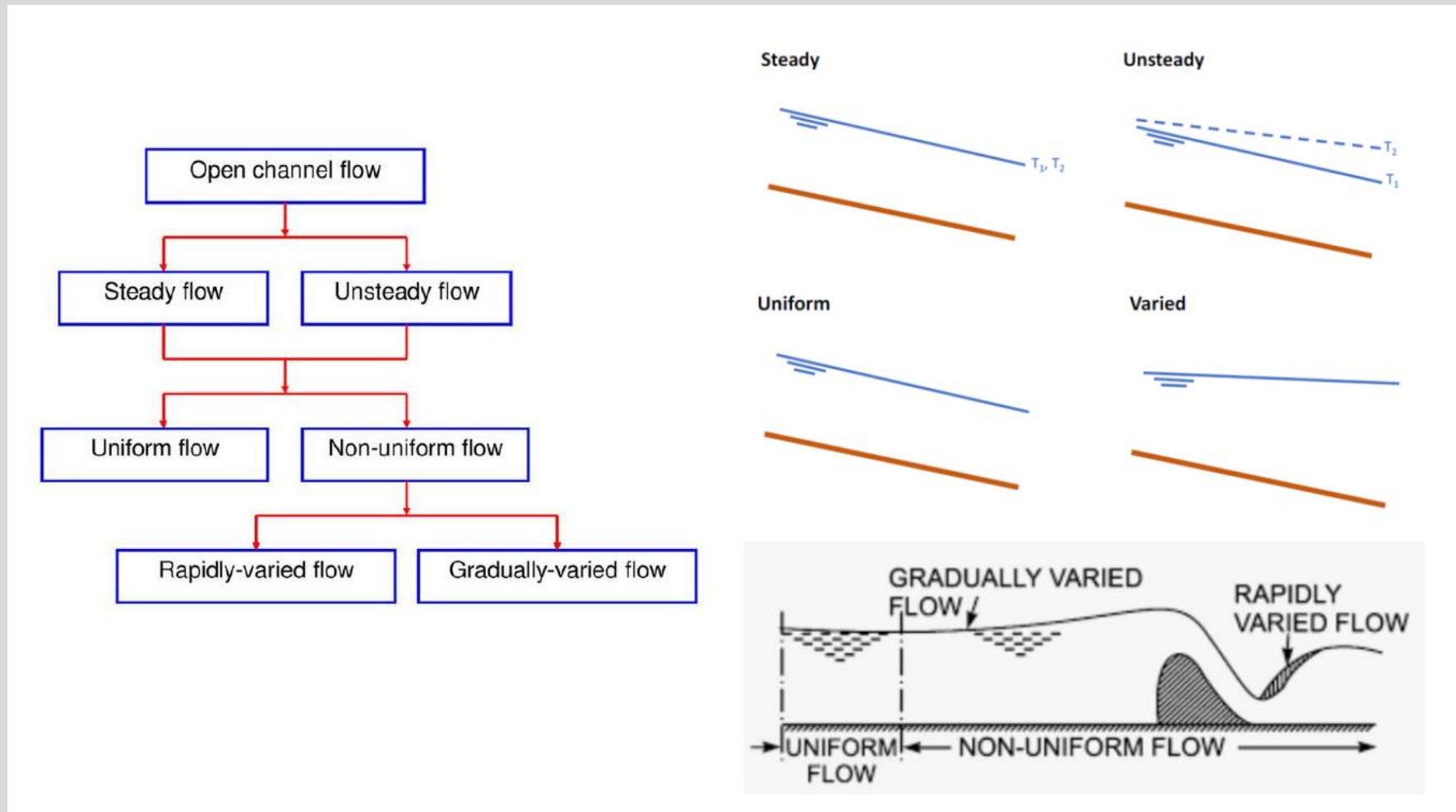
- **flow depth and velocity** in rivers, canals and other conveyance conduits (particularly important during extreme events),
- **changes in flow depth and velocity** due to channel controls, such as weirs, gates and spillways, erosion, construction,
- **dam break** analysis,
- determination of **flood inundation** areas,
- water infrastructure **design** (e.g. reservoirs, bridges, culverts),
- **(bridge) scour**,
- **river restoration**,
- **sediment** transport,
- among many others.

Theoretical Basis for Steady 1D Hydrodynamic Calculations

Review: Classification of Open Channel Flow

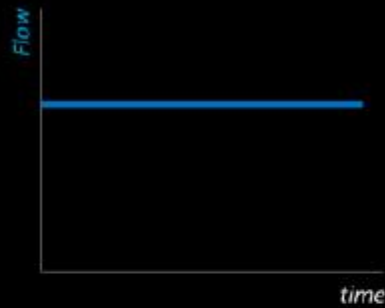


Review: Classification of Open Channel Flow

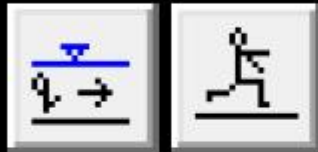


Review: Classification of Open Channel Flow

Steady Flow

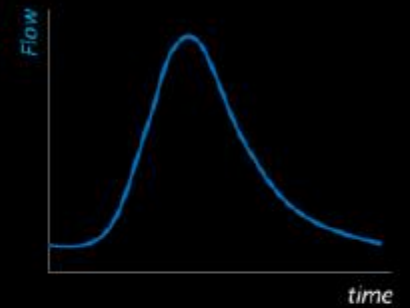


$$Q(t)$$



Unsteady Flow

$$Q(t)$$



1D Steady Flow Water Surface Profiles

- Equations for profile calculations
- Kinetic energy correction factor α

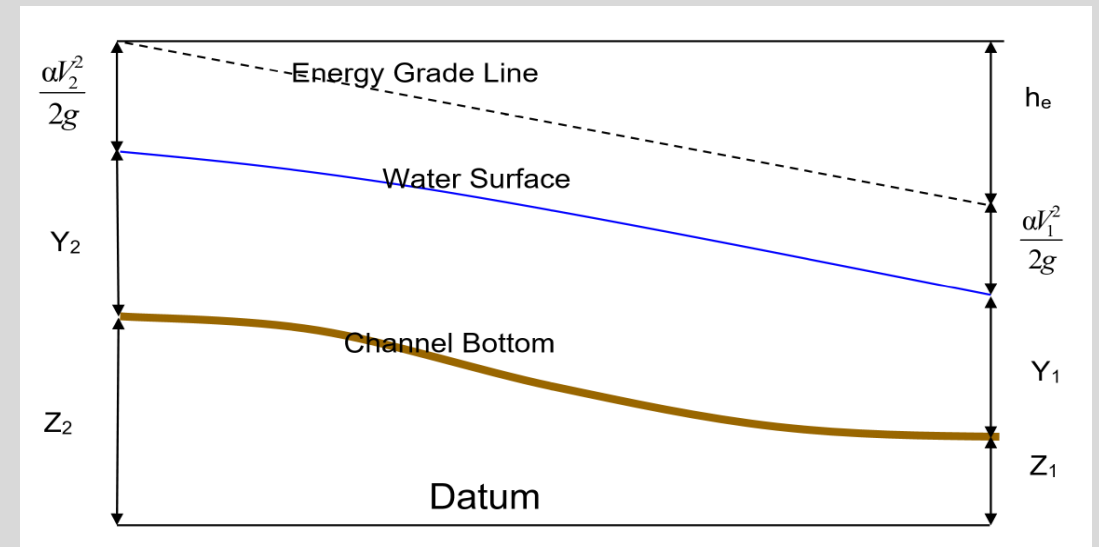
$$Z_2 + Y_2 + \frac{\alpha_2 V_2^2}{2g} = Z_1 + Y_1 + \frac{\alpha_1 V_1^2}{2g} + h_e$$

$$h_e = L\bar{S}_f + C \left| \frac{\alpha_2 V_2^2}{2g} - \frac{\alpha_1 V_1^2}{2g} \right|$$

$$Q = KS_f^{1/2}$$

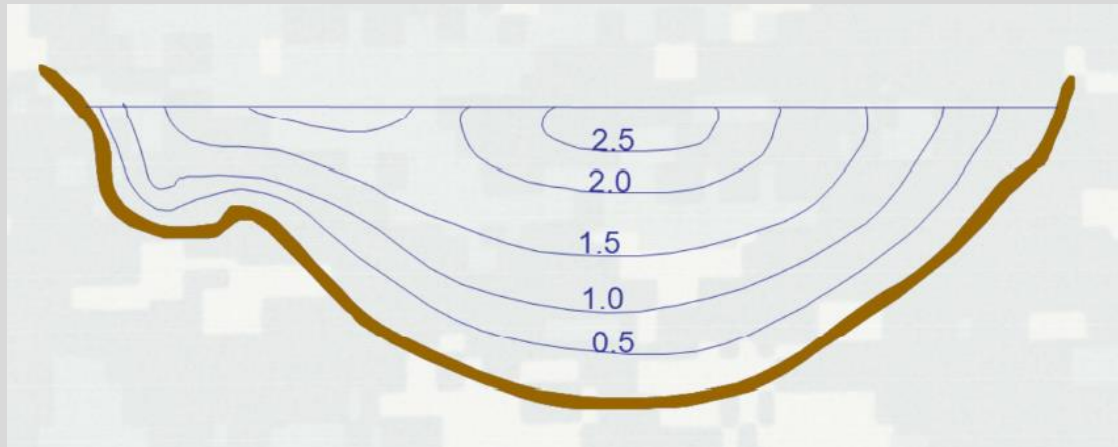
- Conveyance per cross section

$$K = \frac{a}{n} AR^{2/3}$$

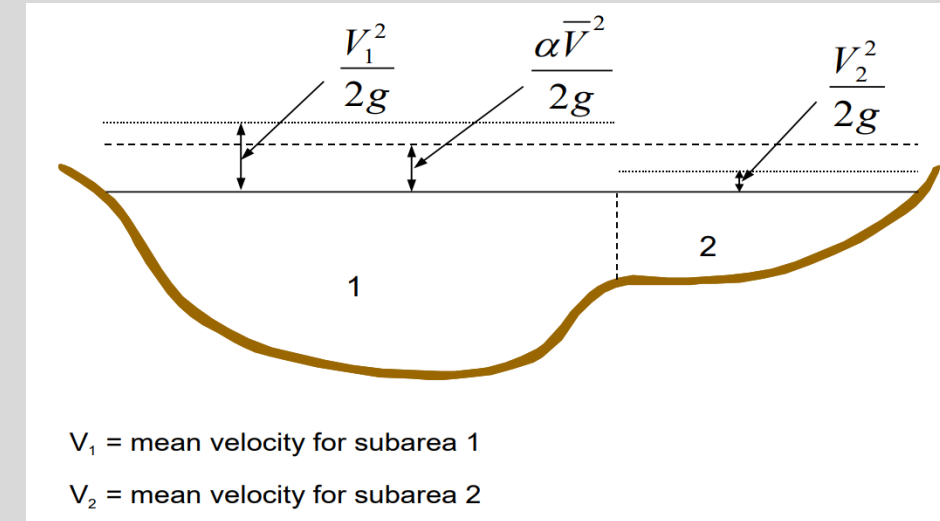


Velocity Distribution & Kinetic Energy Head

- Factors influencing velocity distribution:
 - **Shape** of the cross section
 - **Roughness** of the boundaries
 - Presence of **bends**
 - **Contractions** and **expansions**
 - Flow **obstructions** such as bridge piers



Hydraulic Modelling



$$\alpha \frac{\bar{V}^2}{2g} = \frac{Q_1 \frac{V_1^2}{2g} + Q_2 \frac{V_2^2}{2g}}{Q_1 + Q_2} \quad \alpha = \frac{2g \left[Q_1 \frac{V_1^2}{2g} + Q_2 \frac{V_2^2}{2g} \right]}{\bar{V}^2 (Q_1 + Q_2)}$$

$$\alpha = \frac{Q_1 V_1^2 + Q_2 V_2^2}{\bar{V}^2 (Q_1 + Q_2)}$$

In General:

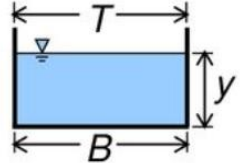
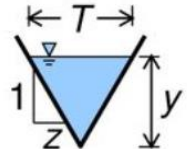
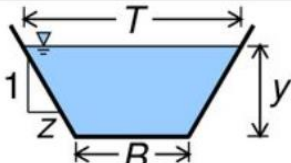
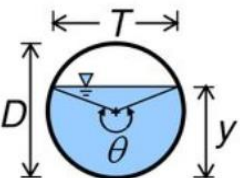
$$\alpha = \frac{[Q_1 V_1^2 + Q_2 V_2^2 + \dots + Q_N V_N^2]}{\bar{V}^2 Q_T}$$

Review: Prismatic and Non-Prismatic Open Channels

- **Prismatic:** constant cross-sectional shape, size, bottom slope
- **Non-prismatic:** varying cross-sectional shape, size, bottom slope

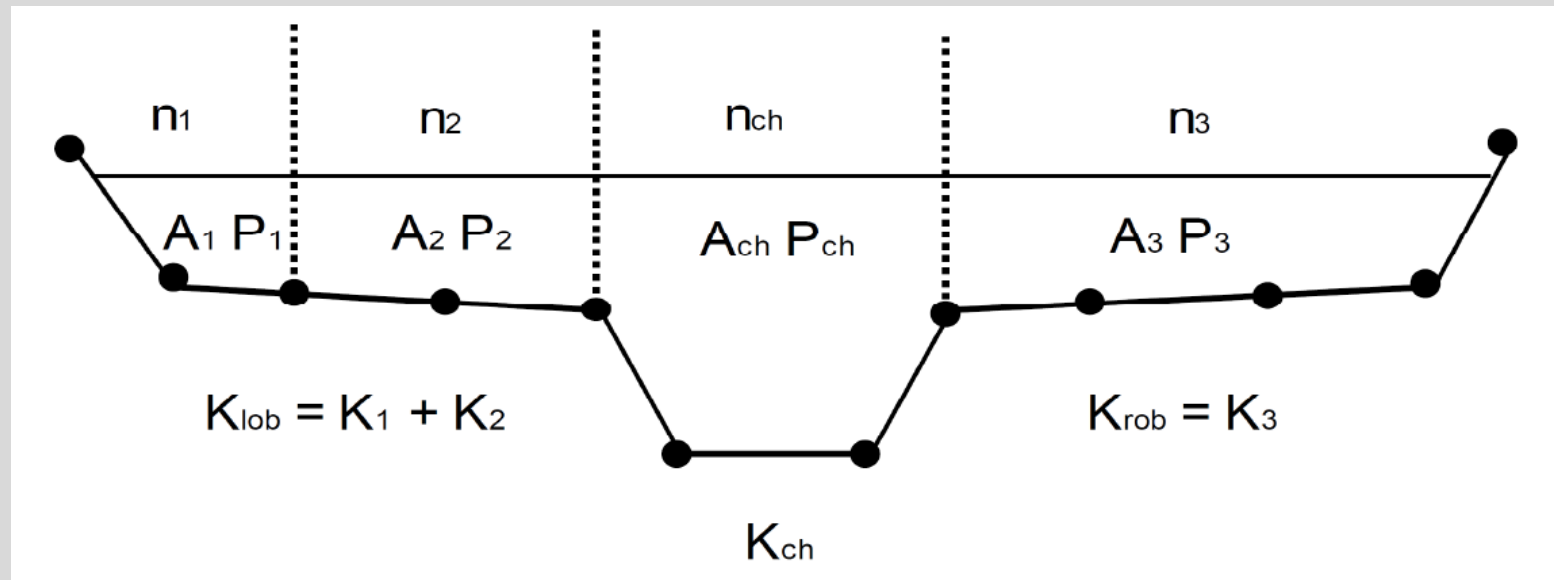


Channel Geometry Parameters

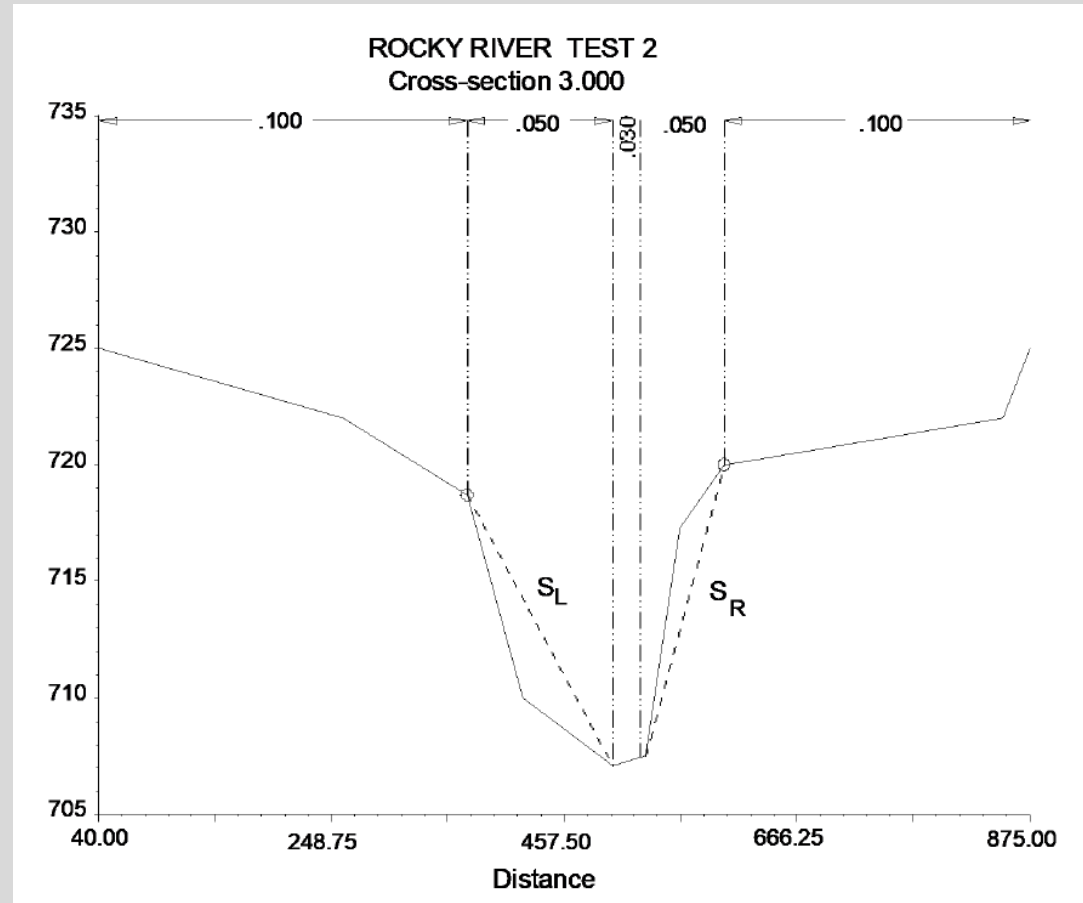
Channel shape	Area A	Top width T	Wetted perimeter P
 Rectangular	By	B	$B + 2y$
 Triangular	zy^2	$2zy$	$2y\sqrt{1+z^2}$
 Trapezoidal	$By + zy^2$	$B + 2zy$	$B + 2y\sqrt{1+z^2}$
 Circle	$\frac{D^2}{8}(\theta - \sin\theta)$ θ in radian	$D\left(\frac{\sin\theta}{2}\right)$ θ in angle	$\frac{\theta D}{2}$ θ in radian

Conveyance in (natural) compound channels

- The **incremental conveyances** in the overbanks are summed up to obtain a conveyance for the left overbank and the right overbank. This is added to the channel flow.



Composite Manning's n for the River



$$n_c = \left[\frac{\sum_{i=1}^N (P_i n_i^{1.5})}{P} \right]^{2/3}$$

Friction Loss Evaluation

Friction slope: **slope of the energy grade line** – examples

- From Manning's equation
- Average conveyance equation
- Other options available.

$$s_f = \left(\frac{Q}{K} \right)^2$$

$$\bar{s}_f = \left(\frac{Q_1 + Q_2}{K_1 + K_2} \right)^2$$

Contraction and Expansion Loss Evaluation

$$h_{ce} = C \left| \frac{\alpha_1 V_1^2}{2g} - \frac{\alpha_2 V_2^2}{2g} \right|$$

- C = contraction or expansion coefficient

Table 3-3 Subcritical Flow Contraction and Expansion Coefficients

	Contraction	Expansion
No transition loss computed	0.0	0.0
Gradual transitions	0.1	0.3
Typical Bridge sections	0.3	0.5
Abrupt transitions	0.6	0.8

Materials for the module on hydraulic modelling

Course reader (available on LEARN).

Lecture slides and notes (available on LEARN before/after).

HEC-RAS software and documentation, available at <https://www.hec.usace.army.mil/software/hec-ras/> (open access).
Installed on computers in various UC computer labs (can be installed via UC 'Software Centre' if not already installed).

Water Surface Profile Computation Procedure in HEC-RAS

The unknown water surface elevation at a cross section is determined by an **iterative solution of the energy equation and the equation for the energy head loss**.

Computational procedure:

1. **Assume a water surface elevation** at the upstream cross section (or downstream cross section if a supercritical profile is being calculated).
2. Based on the assumed water surface elevation, **determine the corresponding total conveyance and velocity head**.

Water Surface Profile Computation Procedure in HEC-RAS

Computational procedure continued:

3. With values from step 2, **compute** $\overline{S_f}$ and **solve the energy head loss equation for** h_e .
4. With values from steps 2 and 3, **solve the energy equation for WS2**.
5. **Compare** the computed value of WS2 with the value assumed in step 1; if required **repeat** steps 1 through 5 until the values agree to within the user-defined tolerance.

Critical Depth Determination

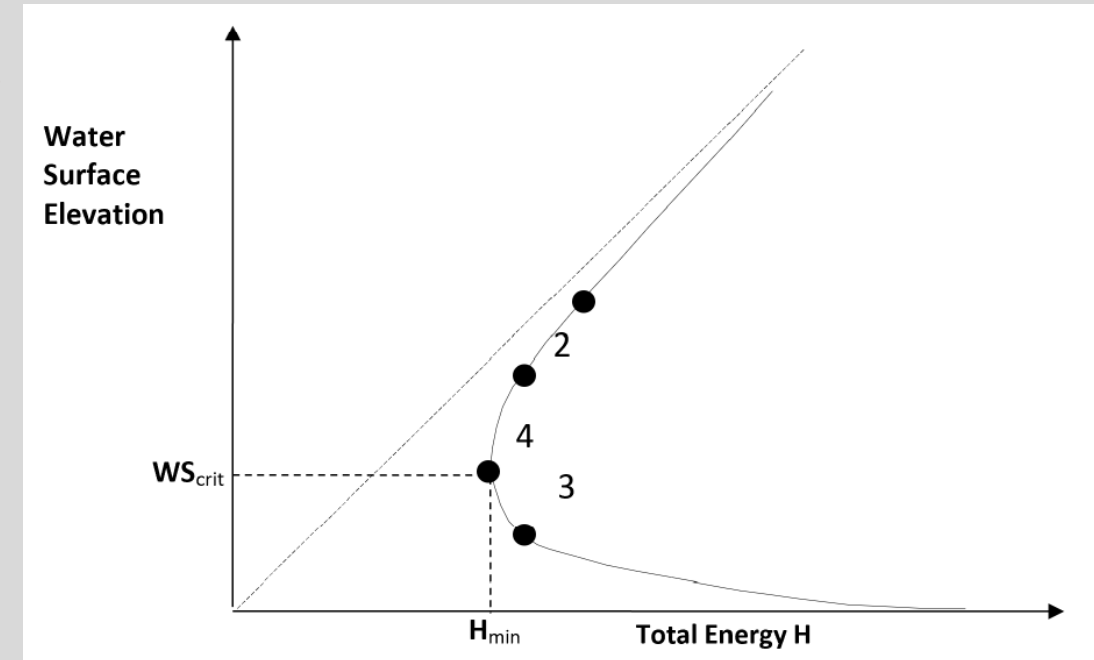
Critical depth for a cross section will be determined if any of the following **conditions** are satisfied:

1. The **supercritical flow regime** has been **specified**.
2. The **calculation of critical depth** has been **requested** by the user.
3. This is an **external boundary cross section** and critical depth must be determined to ensure the user entered boundary condition is in the correct flow regime.
4. The **Froude number check** for a subcritical profile indicates that critical depth needs to be determined to verify the flow regime associated with the balanced elevation.
5. The program could **not balance the energy equation** within the specified tolerance before reaching the maximum number of iterations.

Critical Depth Determination

The critical water surface elevation is the **elevation for which the total energy head is a minimum** (i.e., minimum specific energy for that cross section for the given flow).

The critical elevation is determined with an **iterative** procedure whereby values of **WS** are assumed and corresponding values of **H** are determined with this equation until a **minimum value for H** is reached.



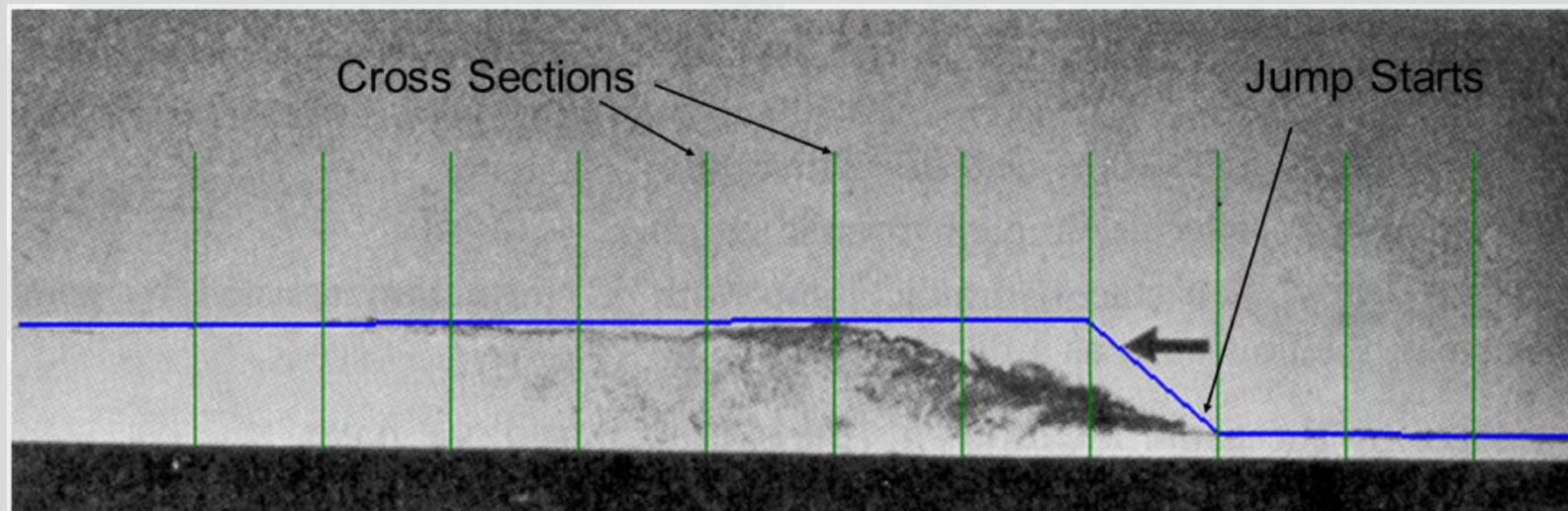
$$H = WS + \frac{aV^2}{2g}$$

$$\frac{Q^2}{g} = \frac{A^3}{T}$$

$$Fr = \frac{V}{\left(\frac{gA}{T}\right)^{1/2}}$$

Mixed Flow Regime

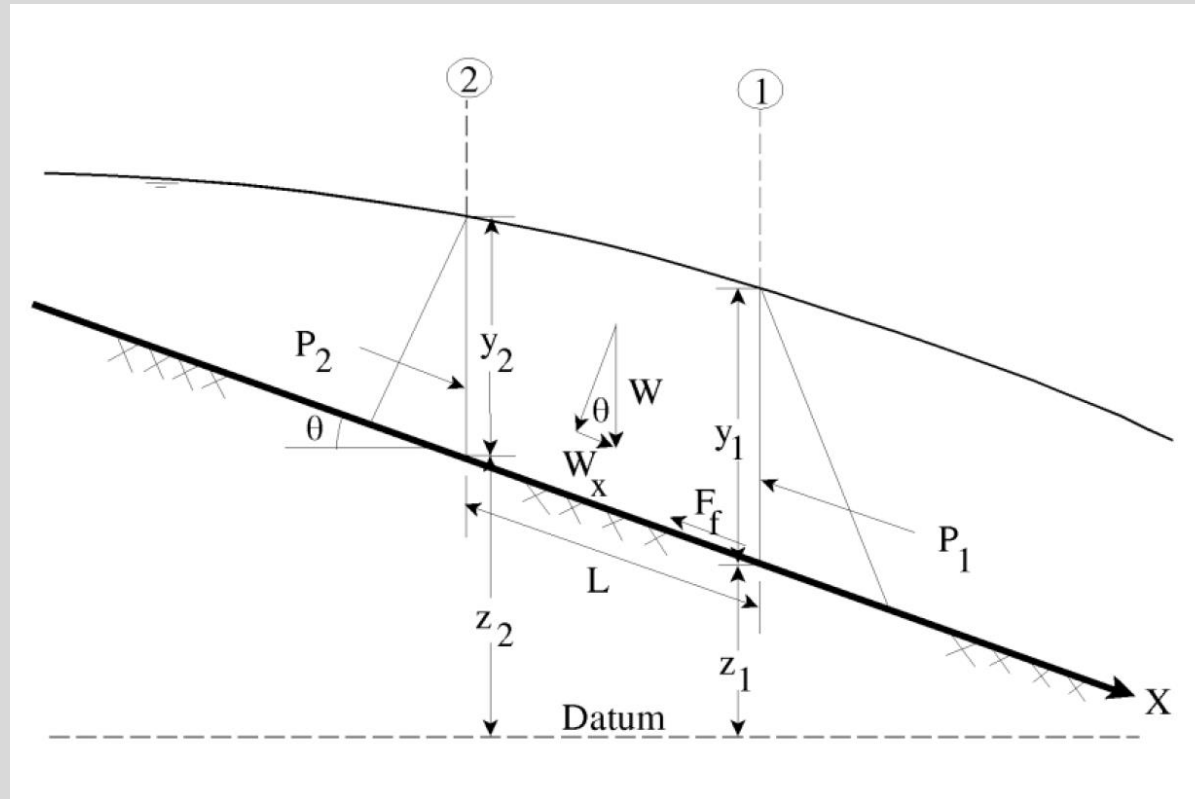
- HEC-RAS will find the cross sections that **bound where the jump starts**
- HEC-RAS does not compute a hydraulic jump length, it **assumes fully developed subcritical flow on the next cross section downstream**



Application of the Momentum Equation

- The energy equation is only applicable to gradually varied flow situations.
- The transition from subcritical to supercritical or supercritical to subcritical is a rapidly varying flow situation.
- **Instances** when the transition from subcritical to supercritical and supercritical to subcritical flow can occur:
 - significant changes in channel slope,
 - bridge constrictions,
 - drop structures and weirs, or
 - stream junctions, among others.
- In **some** of these **instances empirical equations** can be used (such as at drop structures and weirs), while at **others** it is necessary to apply the **momentum equation**.

Momentum Equation

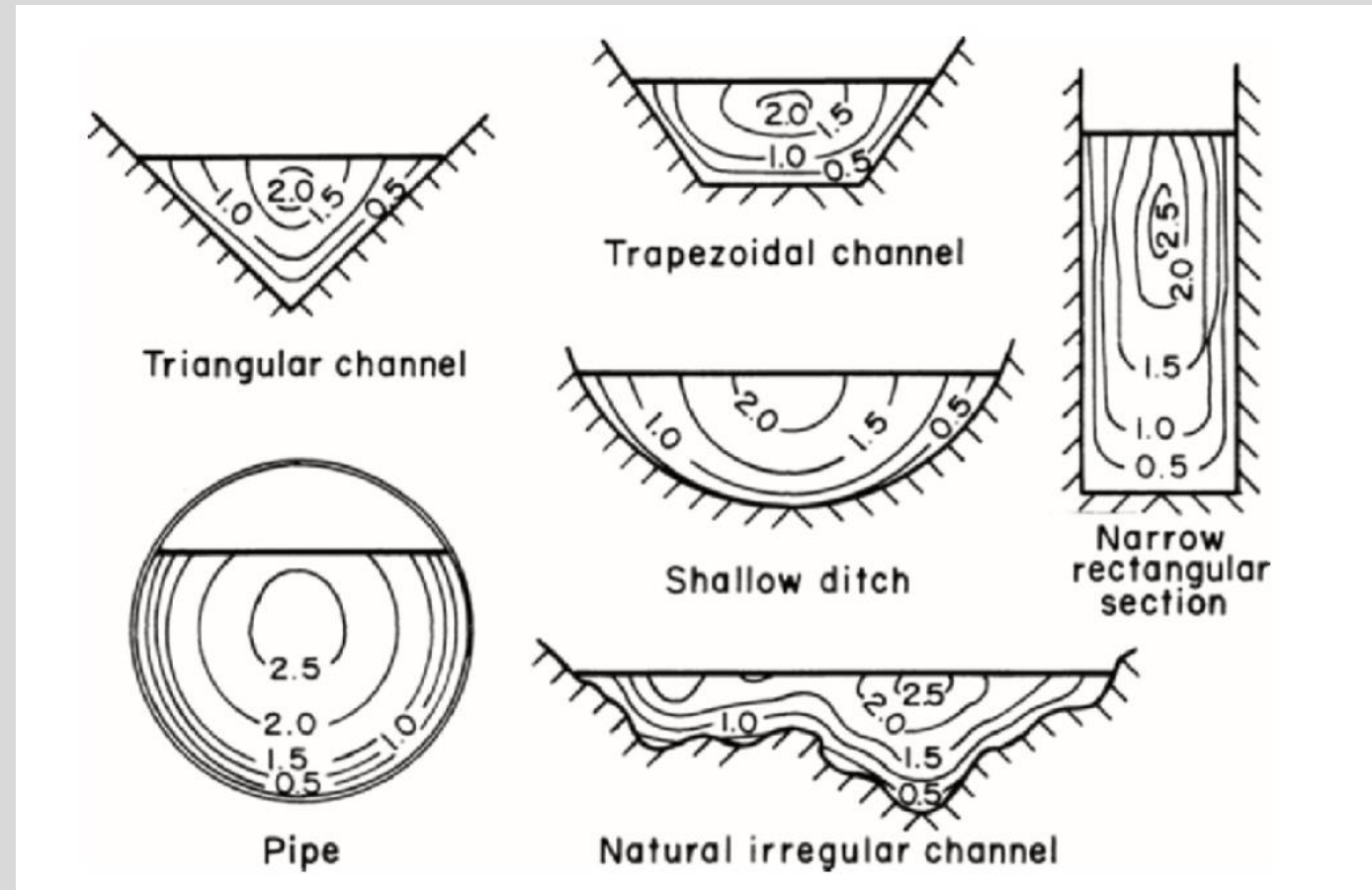


$$\frac{Q_2^2 \beta_2}{g A_2} + A_2 \bar{Y}_2 + \left(\frac{A_1 + A_2}{2} \right) L S_0 - \left(\frac{A_1 + A_2}{2} \right) L \bar{S}_f = \frac{Q_1 \beta_1}{g A_2} + A_1 \bar{Y}_1$$

Momentum Equation

- β : momentum coefficient, that accounts for varying velocity distribution

$$\beta = \frac{\int V^2 dA}{V_m^2 \int dA}$$



Velocity distribution in different channel sections (After Chow [1959])

Values for α and β as given in Chow (1959)

Channel section	α	β
Regular channels	1.10-1.20	1.03-1.07
Natural channels	1.15-1.50	1.05-1.17
Rivers under ice cover	1.20-2.00	1.07-1.33
River valleys, overflowed	1.50-2.00	1.17-1.33

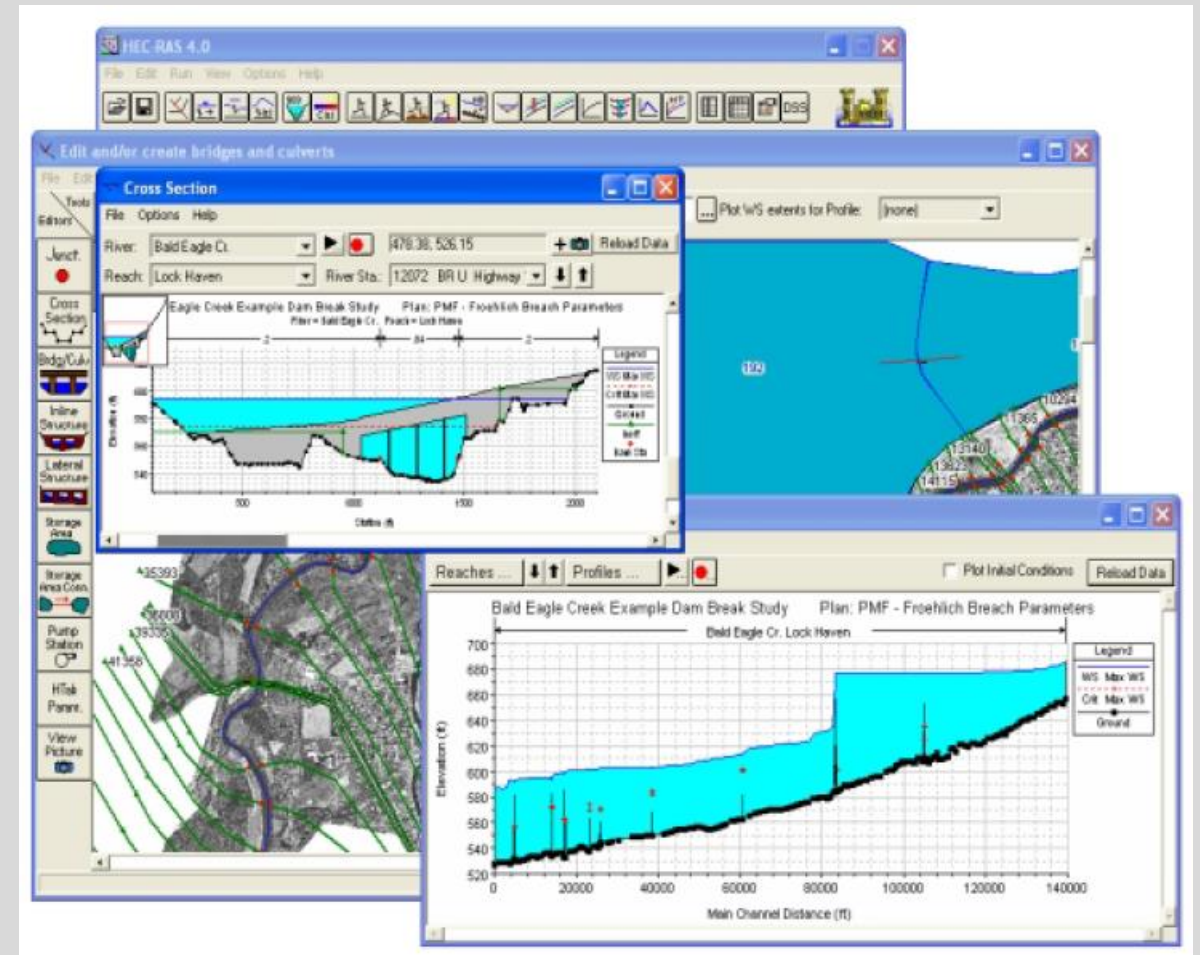
1D Steady HEC-RAS Flow Programme Limitations

The following assumptions are implicit in the analytical expressions used in the current version of the HEC-RAS program:

1. Flow is **steady**.
2. Flow is **gradually varied**. (**Except** at hydraulic structures such as: bridges; culverts; and weirs. At these locations, where the flow can be rapidly varied, the momentum equation or other empirical equations are used.)
3. Flow is **one dimensional** (i.e., velocity components in directions other than the direction of flow are not accounted for).
4. River channels have "**small**" **slopes**, i.e. less than $\sim 1:10$.

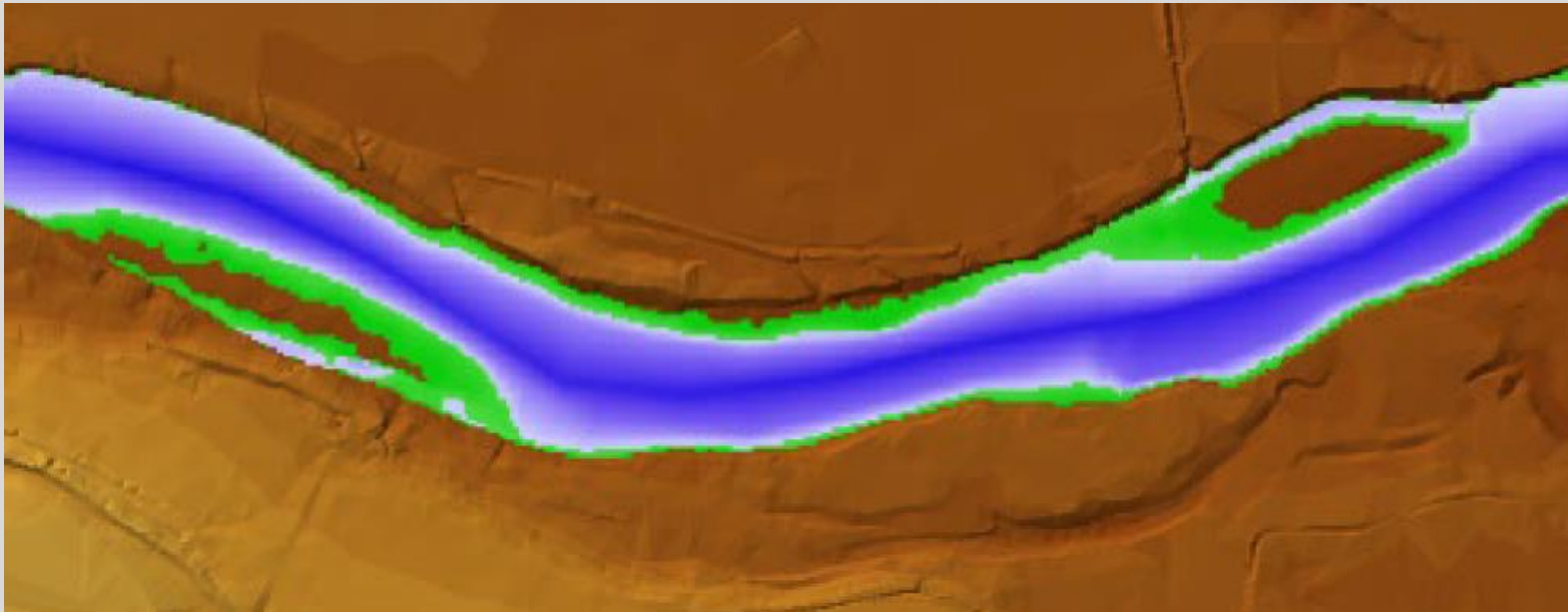
1D vs 2D vs 3D modelling

- **1D** model can be used to determine
 - Depth
 - Velocity
- Cross-sectionally averaged
- Solved typically using a conservation of energy formulation (step-backwater)
- Very sensitive to roughness



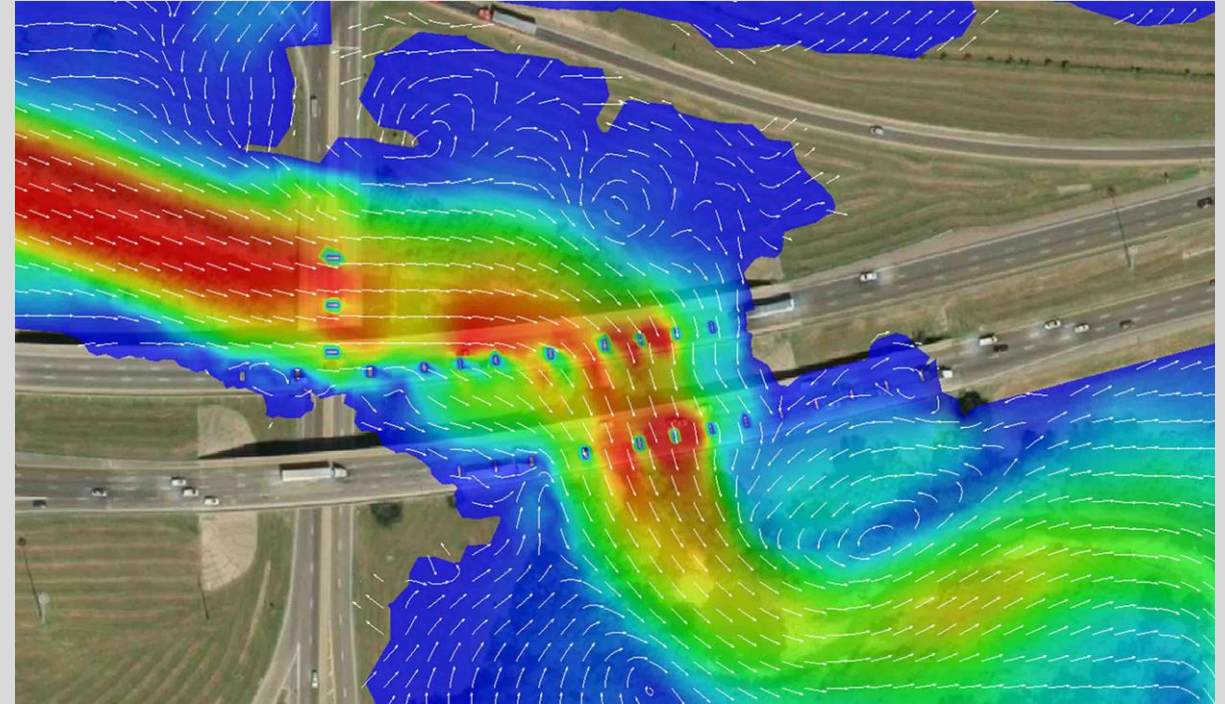
1D vs 2D vs 3D modelling

- **1.5D** model
 - If the topographic surface is available, then the 1D solution can be ,flooded' perpendicular to the channel.



1D vs 2D vs 3D modelling

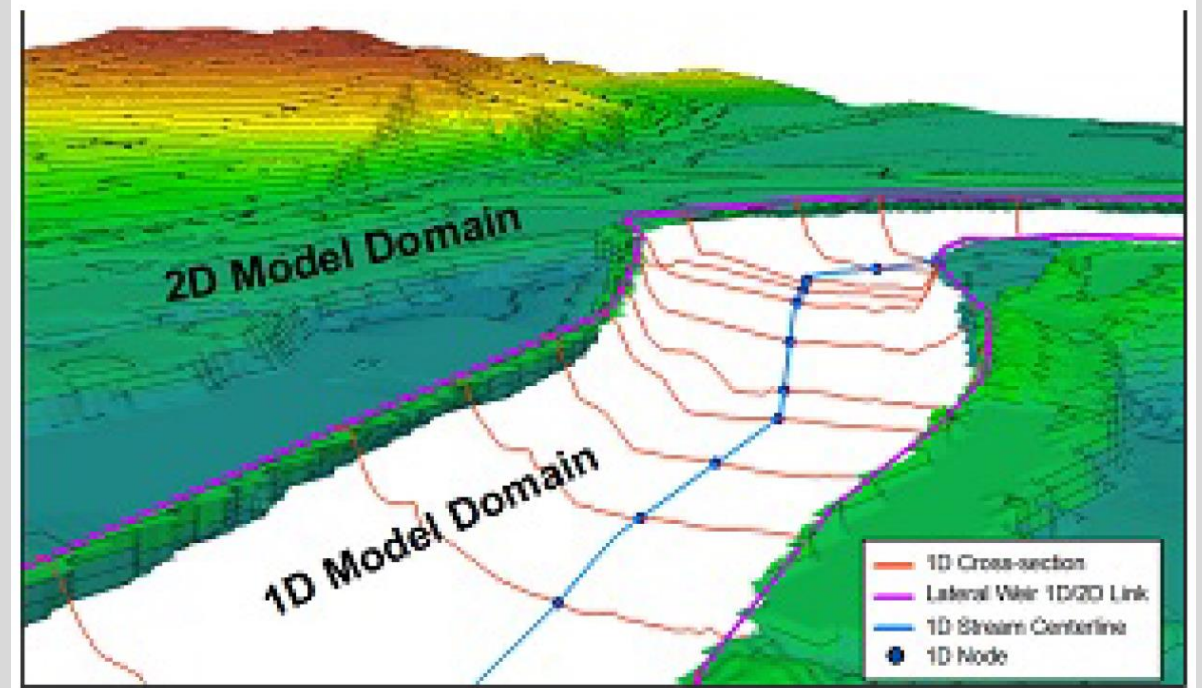
- **2D** model
 - 2D modelling interprets the 2D equations of flow to find out the water depth and depth-averaged velocity on a grid or mesh.
 - It uses digital terrain modelling and bathymetry of a water channel in the question to outline the 2D-flow of the flow.
 - What is 2D modelling? Short video by Dewberry consultants:
<https://www.youtube.com/watch?v=jw19-MOx6aA>



A 2D model inserted over an aerial image of the Lynch Creek bridges. This shows the exact angle and intensity of the flows attacking each pier and pillar. Source: <https://www.stantec.com/en/ideas/completing-the-picture-the-future-of-hydraulic-modeling-is-two-dimensional>

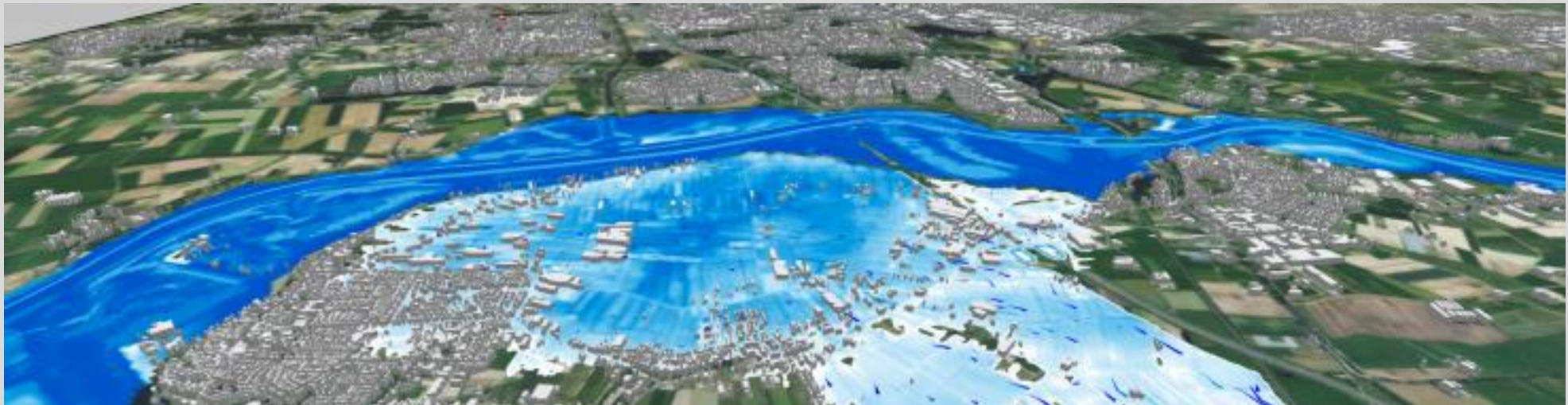
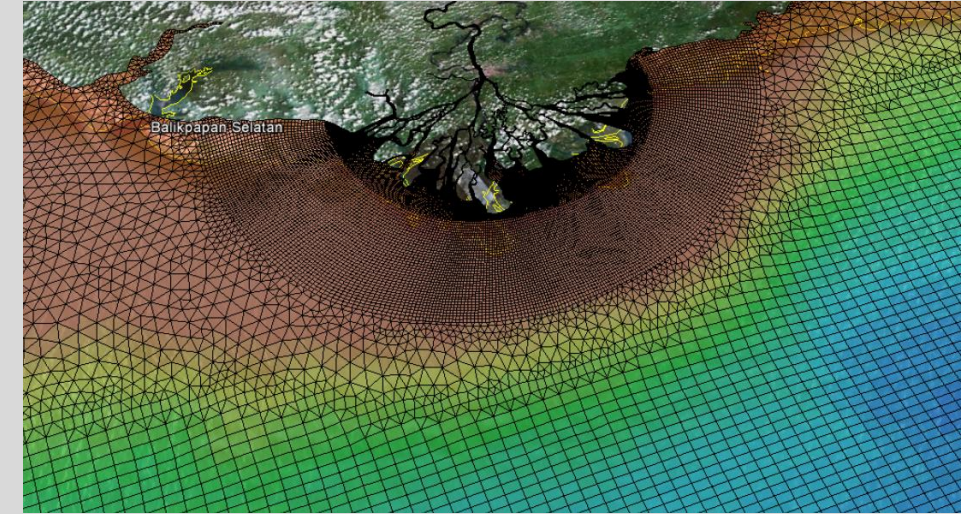
1D vs 2D vs 3D modelling

- **Coupled 1D/2D model**
 - 1D modelling approach in the channel
 - 2D modelling approach in the floodplain
 - 1D/2D coupling at the edge between the two domains



1D vs 2D vs 3D modelling

- **3D** model
 - unsteady shallow water equations
 - some use flexible mesh engine for hydrodynamic simulations on unstructured grids



Classification of Modelling Packages

- According to **what is computed**
 - water surface profiles (HEC-RAS, MIKE)
 - flood waves (DAMBRK, HEC-RAS, MIKE, Flood Modeller)
 - water quality in rivers (QUAL2E)
 - sediment transport (HEC-RAS, MIKE)
 - habitat modelling (PHABSIM)
- **How many dimensions** are used
 - 1D models (HEC-RAS, MIKE11, SOBEK)
 - 2D models (HEC-RAS, Mike21)
 - 3D models (DELFT3D)
- Particulars of the **numerical methods**
 - finite differences
 - finite elements
 - finite volume

To ponder

- **When should steady flow modelling be used and when should unsteady flow modelling be used?**
 - Hints: Cost, Volume, timing, big, quick, perks.
 - In any case: The principle of parsimony applies: the simplest model is the null hypothesis.

Review questions

- What is the kinetic energy correction factor? Provide values for different channel types.
- Explain how the mean kinetic energy head can be calculated for a compound channel.
- What are examples of instances where the flow regime transitions from sub- to supercritical and vice versa? Provide a sketch to support your answer.
- What does the so-called momentum coefficient account for? Give example values for different channel types.

Review questions

- Which equation is solved in 1D HEC-RAS to determine the water surface profile for steady flow conditions? How are losses evaluated? Which alternative equation may be solved for the case of rapidly varied flow?
- Explain the calculation algorithm of the standard step method implemented in HEC-RAS. (5 steps, keep the answer concise)
- For which channel types is the standard step method applicable? Prismatic or non-prismatic? Or both? Explain why.
- Explain what the term conveyance means in the context of hydraulic modelling. In your answer, explain each term of the relevant equation. Can conveyance vary for the main channel and its overbanks? If so, why?

Review questions

- List different 1D steady HEC-RAS flow programme limitations.
- Provide a classification for hydraulic modelling packages.